WORKED SOLUTION

STEP III 1993, Question 3

Pure • LU Matrix Decomposition

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q3

The key idea

Writing $\mathbf{M}$ as a lower-triangular matrix times an upper-triangular one (an *LU decomposition*) turns everything into a chain of one-unknown equations. Multiply out $\mathbf{AB}$ and match entries in a sensible order: each new equation then contains only one unknown, which gives both the values and their uniqueness at once. Triangular matrices are also easy to invert, and $\mathbf{Mx} = \mathbf{v}$ splits into two easy triangular systems.

A warning first. The paper prints $e = 1$, but this is a misprint: the correct value is $e = -1$. With $e = 1$ the bottom row of $\mathbf{AB}$ would be $(3, 10, 1)$, not $(3, 8, 7)$. Everything below uses $e = -1$.

Solution

Finding A and B

Multiplying out,

$$\mathbf{AB} = \begin{pmatrix} a & ap & aq \\ b & bp + c & bq + cr \\ d & dp + e & dq + er + f \end{pmatrix}.$$

We set this equal to $\mathbf{M}$ and take the entries in the following order, so that each equation has just one unknown left in it.

- Entry $(1,1)$: $a = 1$.
- Entries $(1,2)$, $(1,3)$: $ap = 3$ and $aq = 2$, so $p = 3$, $q = 2$ (since $a = 1$).
- Entries $(2,1)$, $(3,1)$: $b = 4$, $d = 3$.
- Entry $(2,2)$: $bp + c = 13$, so $c = 13 - 12 = 1$.
- Entry $(2,3)$: $bq + cr = 5$, so $8 + r = 5$ and $r = -3$ (using $c = 1$).
- Entry $(3,2)$: $dp + e = 8$, so $9 + e = 8$ and $e = -1$.

- Entry $(3,3)$: $dq + er + f = 7$, so $6 + 3 + f = 7$ and $f = -2$.

Every unknown was *forced* by an equation in which everything else was already known, and the only divisions were by $a = 1$ and $c = 1$, which are non-zero. So if $\mathbf{M} = \mathbf{AB}$ then the unknowns must take these values. Conversely, these values satisfy all nine equations. So the solution exists and is unique:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -1 & -2 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix},$$

with $a = 1$, $b = 4$, $c = 1$, $d = 3$, $e = -1$, $f = -2$, $p = 3$, $q = 2$, $r = -3$.

Ask yourself Where does uniqueness come from? We only found one solution.

Uniqueness comes from the *order* of the deductions. No choice was made at any stage. Each equation was linear in a single new unknown with a non-zero coefficient, so it had exactly one solution. Any $\mathbf{A}$, $\mathbf{B}$ of the given forms with $\mathbf{AB} = \mathbf{M}$ must therefore be the pair above. (The 1s on the diagonal of $\mathbf{B}$ matter: without them we could scale $\mathbf{A}$ up and $\mathbf{B}$ down.)

The inverses

To invert $\mathbf{A}$, we solve $\mathbf{Au} = \mathbf{v}$ from the top down:

$$\begin{aligned} u_1 &= v_1, \\ u_2 &= v_2 - 4u_1 = -4v_1 + v_2, \\ -2u_3 &= v_3 - 3u_1 + u_2 = -7v_1 + v_2 + v_3 \implies u_3 = \tfrac{1}{2}(7v_1 - v_2 - v_3). \end{aligned}$$

To invert $\mathbf{B}$, we solve $\mathbf{Bw} = \mathbf{u}$ from the bottom up:

$$\begin{aligned} w_3 &= u_3, \\ w_2 &= u_2 + 3w_3 = u_2 + 3u_3, \\ w_1 &= u_1 - 3w_2 - 2w_3 = u_1 - 3u_2 - 11u_3. \end{aligned}$$

Reading off the coefficients,

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ \frac{7}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \qquad \mathbf{B}^{-1} = \begin{pmatrix} 1 & -3 & -11 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplying out confirms that $\mathbf{AA}^{-1} = \mathbf{I}$ and $\mathbf{BB}^{-1} = \mathbf{I}$. Notice that the inverse of a lower (upper) triangular matrix is again lower (upper) triangular.

The simultaneous equations

The equations say $\mathbf{Mx} = \mathbf{v}$ with $\mathbf{x} = (x, y, z)^{\mathrm{T}}$ and $\mathbf{v} = (7, 18, 25)^{\mathrm{T}}$. Since $\mathbf{M} = \mathbf{AB}$,

$$\mathbf{x} = \mathbf{M}^{-1}\mathbf{v} = \mathbf{B}^{-1}\mathbf{A}^{-1}\mathbf{v}.$$

Note the order: $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$. First,

$$\mathbf{u} = \mathbf{A}^{-1}\mathbf{v} = \begin{pmatrix} 7 \\ -28+18 \\ \frac{1}{2}(49-18-25) \end{pmatrix} = \begin{pmatrix} 7 \\ -10 \\ 3 \end{pmatrix},$$

and then

$$\mathbf{x} = \mathbf{B}^{-1}\mathbf{u} = \begin{pmatrix} 7+30-33 \\ -10+9 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}.$$

So the unique solution is

$$x = 4, \qquad y = -1, \qquad z = 3.$$

It is unique because $\mathbf{M}$ is invertible: $\det \mathbf{M} = \det \mathbf{A} \det \mathbf{B} = (1 \cdot 1 \cdot (-2))(1) = -2 \neq 0$.

Checks

Substitute back: $4-3+6=7$, $16-13+15=18$, $12-8+21=25$. All three equations hold. We also checked with exact fractions that $\mathbf{AB} = \mathbf{M}$ and that $\mathbf{B}^{-1}\mathbf{A}^{-1}$ really is $\mathbf{M}^{-1}$; its bottom row is $(\frac{7}{2}, -\frac{1}{2}, -\frac{1}{2})$. With the printed value $e = 1$, the product $\mathbf{AB}$ has bottom row $(3, 10, 1)$, which confirms the misprint.

Common slips

- Trusting the printed $e = 1$ and then forcing the $(3,2)$ and $(3,3)$ entries to fit. Always multiply your $\mathbf{A}$ and $\mathbf{B}$ back together.
- Using $\mathbf{A}^{-1}\mathbf{B}^{-1}$ instead of $\mathbf{B}^{-1}\mathbf{A}^{-1}$.
- Showing that the values work without arguing that no other values could.

Beyond STEP

This is the *LU decomposition*, which is how computers actually solve linear systems. Gaussian elimination is really the process of factorising $\mathbf{M} = \mathbf{LU}$. After that, each new right-hand side costs only a forward substitution and a back substitution, with no fresh elimination. In general a row swap may be needed first, which gives $\mathbf{PM} = \mathbf{LU}$. Strang's *Introduction to Linear Algebra* builds a whole chapter around this idea.

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